Mating Connector:
Housing: Molex 0436450200
Precrimped 18AWG wire 450mm:
Black: Molex 2147601225
Red: Molex 2147602225

Reverse Polarity Protection

Q14
CSD19534Q5A

J10
POWER
V_IN
V_IN_RTN
Molex
0436500200

D1
SMDJ24CA
24V

C64
10uF-50V-10%-X7R-1210

C62
0.1uF-50V-10%-0603

U11
LM74700QDBVRQ1
VCAP
EN
GND
A
G
C

C55
10uF-50V-10%-X7R-1210

R29
5.1k-1%-0603

R31
5.1k-1%-0603

R25
5.1k-1%-0603

D2
PWR

VIN_PS

File: buck-boost-power-supply.kicad_sch
2/5
+V_IN
-V_IN
15V 3A Buck-Boost Power Supply
+V_OUT
-V_OUT

Power Supply Back
Feed Protection

Q1
CSD19534Q5A

C1
10uF-50V-10%-X7R-1210

C15
0.1uF-50V-10%-0603

U2
LM74700QDBVRQ1
VCAP
EN
GND
A
G
C

C16
10uF-50V-10%-X7R-1210

J5
V_OUT
V_OUT_RTN
15V
Molex
0436500200

Mating Connector:
Housing: Molex 0436450200
Precrimped 18AWG wire 450mm:
Black: Molex 2147601225
Red: Molex 2147602225

Mating Connector:
Housing: Molex 5040510401
Precrimped 24AWG wire 150mm: Molex 0797581003

J7
USB1
Molex
5040500491

File: canable2.kicad_sch
3/5
CANable2.0
USB_DP
USB_DM
3V3_USB
GND_CAN
CAN_HC
CAN_LC

J9
CAN1
MFG P/N: 0705550036
MFG: Molex

J1
Terminal

J2
Terminal
Keystone Electronics
7774

File: NX-Button-Interface.kicad_sch
4/5
NX-Button Interface

File: usb-serial.kicad_sch
5/5
USB Serial
+5V J401
3V3 J401
RXD
TXD
RESET
FORCE_RECOVERY

PH1RB-15-UA
Adam Tech
15 3V3_BTN
14
13
12 NX_PC_LED-
11 VDD_5V_SYS
10 UART2_RXD
9 UART2_TXD
8 AC OK
7 Auto Power On Disable
6
5 SYS_RESET
4
3 REC
2 GND
1 SLEEP/WAKE*
NX CONTROL & UART
J3
GND1

VDD_5V_SYS
3V3_BTN
UART2_TXD
UART2_RXD
SYS_RESET
REC
VDD_5V_SYS
NX_PC_LED-

D3
PC

SW2
RECOVERY

SW1
RESET

GND1

Light Pipe
LP1
Bivar
Bivar-SLP3-150-100-F

Light Pipe
LP2
Bivar
Bivar-SLP3-150-100-F

Light Pipe
LP3
Bivar
Bivar-SLP3-150-100-F

Light Pipe
LP4
Bivar
Bivar-SLP3-150-100-F

H3
MountingHole
H10
MountingHole
H11
MountingHole
H4
MountingHole

H5
MountingHole
H7
MountingHole
H8
MountingHole

H1
1.152mm ToolingHole
H12
1.152mm ToolingHole
H2
1.152mm ToolingHole
H13
1.152mm ToolingHole

FID6 Fiducial
FID1 Fiducial
FID3 Fiducial
FID5 Fiducial
FID4 Fiducial
FID2 Fiducial

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Sheet: /
File: TVS-Adapter-v1.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-11-01	Rev: 1
KiCad E.D.A. 10.0.8		Id: 1/5

+V_IN
−V_IN
+V_OUT
−V_OUT
GND
C8 47uF 50V
C12 47uF 50V
10uF-50V-10%-X7R-1210
C54
C10
C9
C7
C11
C53
C59
C60
C6
C19
C24
C23
C18
C29
C28
C2
C4
C5
C3
HDRV1
LDRV1
Q13 CSD19534Q5A
Q10 CSD19534Q5A
Q7 CSD17578Q5A
Q9 CSD17578Q5A
HDRV2
LDRV2
SW1
SW2
L1
2.2uH-19A-5.1mohm-20%
RCS
CSG
R18 10mOhm-3W-1%-2512
D8 BAT54KFILM
R30 249K-1%-0603
R27 93.1k-1%-0603
R23 2K-1%-0603
R28 10R0-1%-0603
C47 0.1uF-50V-10%-0603
U7
Texas Instruments
LM5176PWPR
VIN
VISNS
EN/UVLO
SS
COMP
MODE
RT
DITH
SLOPE
CS
CSG
VCC
BIAS
VOSNS
HDRV1
BOOT1
SW1
LDRV1
FB
PGOOD
HDRV2
BOOT2
SW2
LDRV2
ISNS(+)
ISNS(−)
AGND
PGND
EP
C45 0.1uF-50V-10%-0603
R32 0R-1%-0603
R24 0R-1%-0603
R15 0R-1%-0603
R17 0R-1%-0603
D7 BAT54KFILM
D6 BAT54KFILM
C48 0.1uF-50V-10%-0603
C35 0.1uF-50V-10%-0603
C44 1uF-50V-10%-0603
Vcc
R7 357K-1%-0603
R8 20K-1%-0603
15nF-10%-50V-0603 C39
5.6nF-10%-50V-0603 C31
C34
5.6nF-10%-50V-0603 C36
5.9k-1%-0603 R9
270pF-10%-50V-0603
93.1k-1%-0603 R21
13K-1%-0603 R19
150pF-50V-10%-0603 C41
C32
100R-1%-0603 R12
47pF-50V-10%-0603
R10 100R-1%-0603

VIN(MAX)	24V
VIN(MIN)	6V / 5V Transient
VOUT	15V
IOUT(MAX)	3A
FSW	589 kHz
Efficiency	96.7%

15V-3A-Power Supply Analysis

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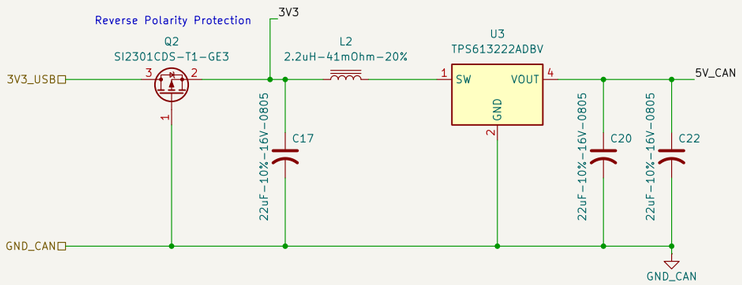
Sheet: /15V 3A Buck-Boost Power Supply/
File: buck-boost-power-supply.kicad_sch

Title: The Vision System Adapter

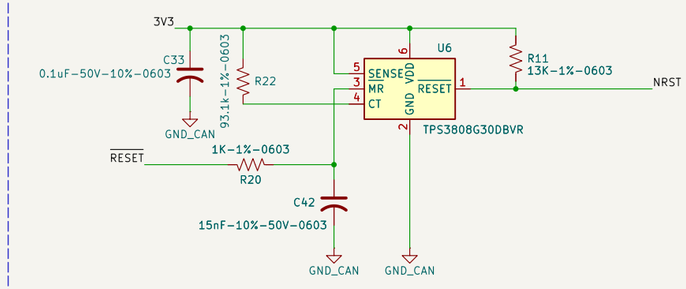
Size: B	Date: 2026-11-01	Rev: 1
KiCad E.D.A. 10.0.8		Id: 2/5

5V-250mA-Power Supply Analysis

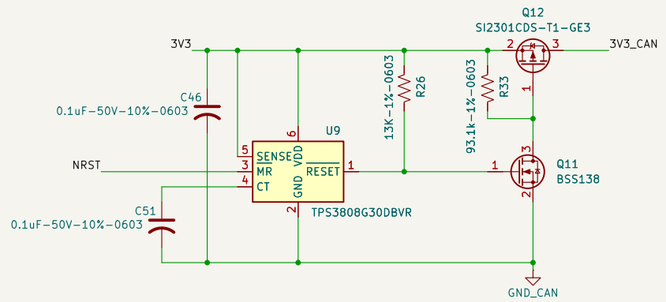
Reverse Polarity Protection

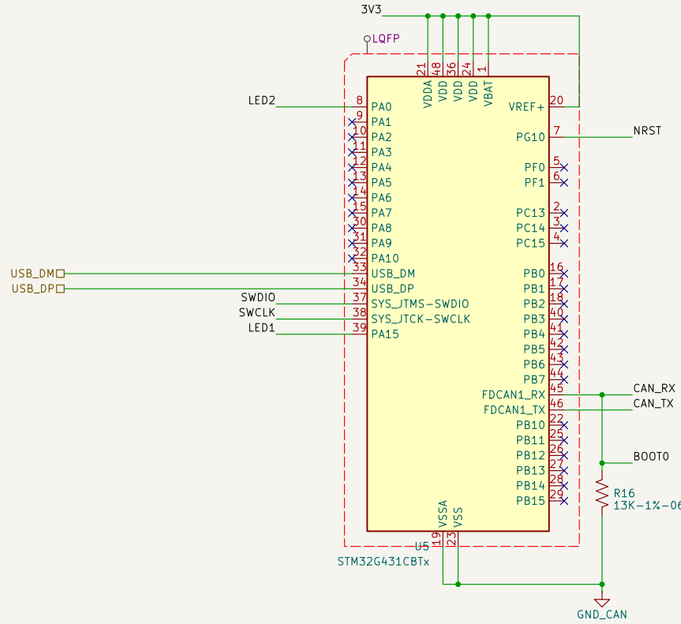


POWER ON RESET
300ms

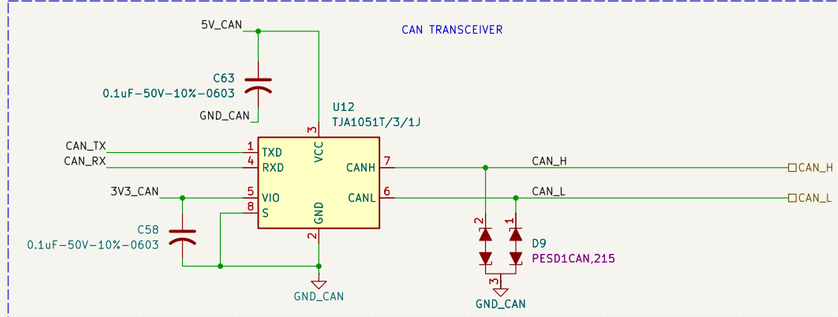


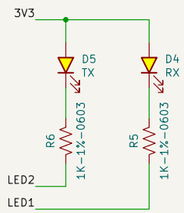
CAN BOOTO POWER DELAY
0.5 seconds

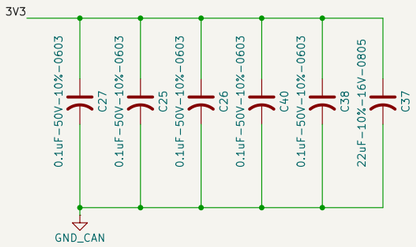




CAN TRANSCEIVER







DEBUG HEADER



NOTES:
- CANABLE circuit interfaces with the M.2 Key-E adapter plugged into the J401. Input power must be no greater than 3.6V

CANable 2.0 Schematic

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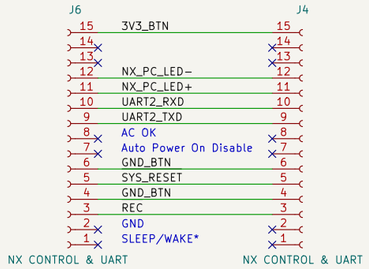
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Sheet: /CANable2.0/
File: canable2.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.8		Id: 3/5



H6
M3.7 MountingHole

H9
M3.7 MountingHole

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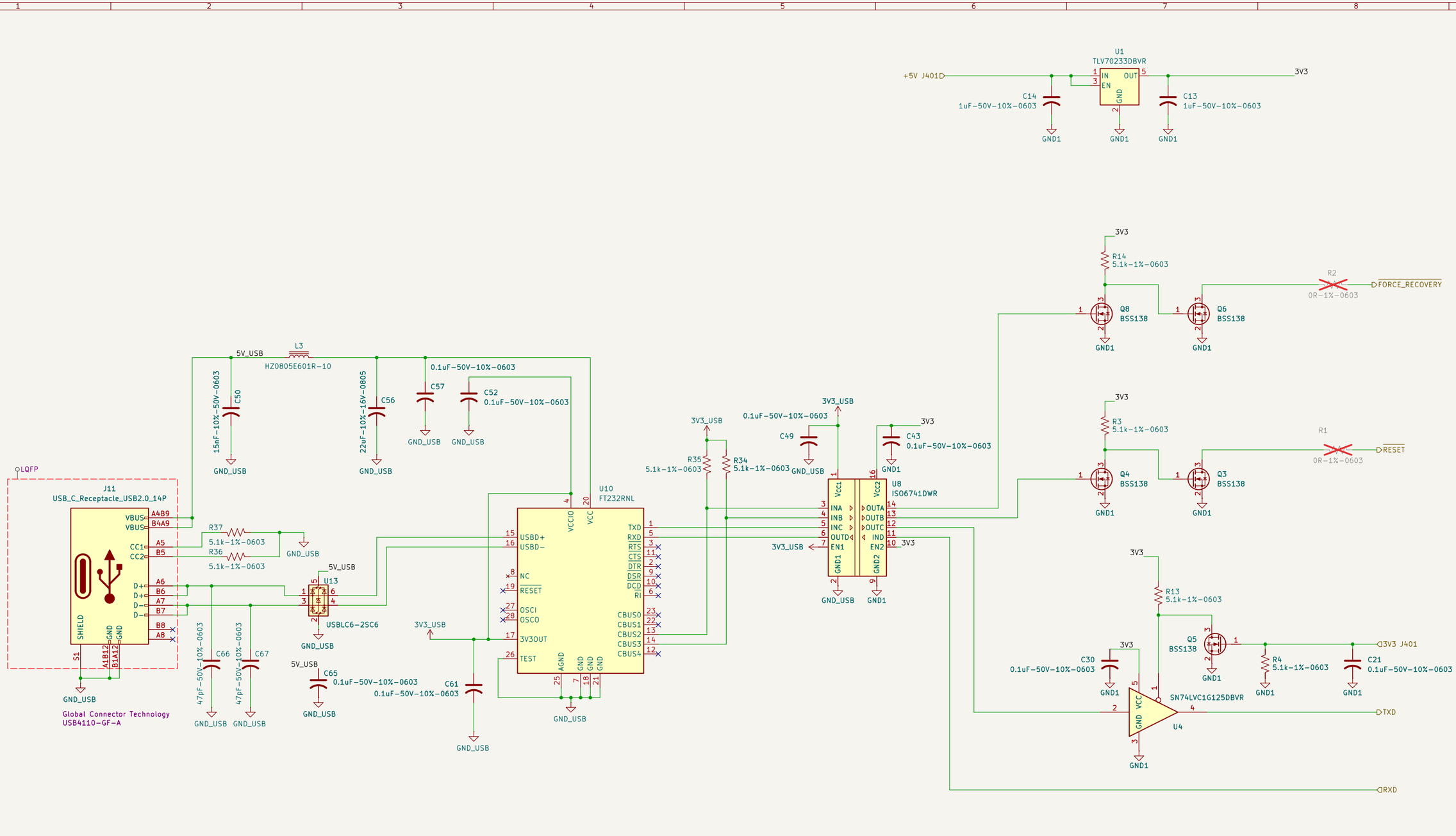
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Sheet: /NX-Button Interface/
File: NX-Button-Interface.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.8		Id: 4/5



https://hackaday.com/2023/08/07/all-about-usb-c-example-circuits/

NOTE: Program CBUS2 and CBUS3 to CBitBangI/O with FT_PROG or ftdi_eeprom before installing R1/R2.

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Sheet: /USB Serial/
File: usb-serial.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-11-01	**Rev: 1**
KiCad E.D.A. 10.0.8		Id: 5/5